Data structures and Algorithms

Week 4-5

I chose the Binary Search simply because of complexity, I felt it was harder to write and wanted to challenge myself. That same reason was also why I did both DFS and BFS searches on the Data Tree structure. Using the different searches I found four suspects.

Helena Blackwood, ID: 764, Entered: 22:13:17, Never left

Nem Wellington, ID: 288, Entered: 22:13:17, Never left

Alfred Pugsley, ID: 192, Entered: 22:13:17, Exited: 22:14:54

Jasper Wellington, ID: 47, Entered: 22:13:17, Never left

Of those four only one person matches what we know about the murderer both entering and exiting between 22:00-23:00, thus I conclude that the murderer is Alfred Pugsley.

The reason I chose a tree over a hash map was I felt it was easier to understand and felt it better fit the task.

Challenges

Most challenges I have had during this course come mainly from struggling to understand the things we were working on that week or what exactly the assignment that week was. I think the hardest one to implement was either my merge sort or my Binary tree, more specifically getting the BFS and DFS functions (which was not the entire issue. There were also issues with my ‘Insert’ function) in the search tree to work. After lots of debugging I saw the issues, stuff like missed lines, or misunderstood pseudo code. But oftentimes it was just me not understanding how stuff worked.

Overall, I have had a challenging but rewarding time with this course, I do wish we would have had this course alongside the C# course however as I feel it could have been useful for the understanding of both.

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